

Detailed testing of the Hawaiian Koa (*Acacia koa*)

The Hawaiian koa is an endemic tree species of high ecological (e.g., provides key habitat for many threatened species and watershed recharge areas), economical (e.g., highly regarded hardwood timber) and cultural importance (e.g., timber is used to build traditional Hawaiian outrigger canoes for fishing, racing, and voyaging) (Baker et al., 2009; Mitchell et al., 2005). It is distributed on the four main Hawaiian Islands (the Island of Hawai'i, Maui, O'ahu and Kaua'i). Hawaiian koa grows in various habitats ranging from wet to drier areas in different elevations and therefore also expresses high variation in phenotypic traits (Baker et al., 2009). It is currently listed under the threat category 'least concern' in the IUCN Red list; however, the last assessment of koa was in 2010 (Contu, 2010). In 2015 an action plan for Hawaiian koa was developed to discuss the actions needed to achieve healthy koa forests that accommodate people and economic needs and promote forest conservation (Inman-Narahari, 2015).

Please refer to a detailed description of evidence used for the Hawaiian koa to DOI: 10.1093/biosci/biag042.

1. Fill in species and scenario information.

We tested four different candidate scenarios: (1) a single metapopulation (null hypothesis), (2) two subpopulations based on genetic data (microsatellite markers; Fredua-Agyeman et al., 2008), (3) seven potential subpopulations, based on genomic data (genotyping-by-sequencing GBS; Gugger et al., 2018) and (4) four subpopulation based the geographic distribution on the four main Hawaiian oceanic islands.

Phase 1: Identify Subpopulations

This survey is best views on Desktop, or rotate your phone for a better view.

Please use the dropdown menu below for additional information.

- Subpopulations
- Evolutionarily Significant Units (ESU)
- Possible Evolutionarily Significant Unit (pESU)
- Gene Flow
- Potential IUCN Species assessment
- References

Scoring:

- 2 points or higher = distinct subpopulations
- Less than 2 points = stop assessment

Lines of evidence

genetic

Which species are you testing? *

Acacia koa

How to set up scenarios to test:

This survey allows the evaluation of **multiple subpopulation scenarios**.

The **first scenario always represents the null hypothesis**, which assumes a **single metapopulation**. This one-metapopulation model must be tested first and serves as the baseline against which all alternative scenarios involving multiple subpopulations are compared.

When selecting scenarios that include **multiple subpopulations** (i.e. not the default metapopulation), **all subpopulations must be mutually distinct and supported by evidence**. Each subpopulation must have a clear evidentiary basis for inclusion in the scenario being tested. If any two subpopulations cannot be distinguished from one another, they should be merged, and a revised scenario should be considered for testing.

How many Scenarios do you want to assess?

4 Scenarios

1 Metapopulation	2 Subpopulations	7 Subpopulations	4 Subpopulations
Scenario 1 name	Scenario 2 name	Scenario 3 name	Scenario 4 name

2. Answer the questions for each scenario.

a. Genetic evidence

In **Scenario (1)** (null hypothesis), depicting a single metapopulation. For this scenario scoring is reserved to keep the questions identical across all scenarios. The existing evidence did not indicate the presence of a single metapopulation of the Hawaiian koa. Therefore, a score of 0/2 for genetic structure was assigned.

In **Scenario (2)**, a genetic study identified two subpopulations (Kaua'i vs. Island of Hawai'i, Maui, and O'ahu) based on clustering analyses and pairwise F_{ST} values (Fredua-Agyeman et al., 2008). This resulted in a score of 2/2 for genetic structure.

In **Scenario (3)**, a genomic study identified seven subpopulations (one each on Kaua'i, Maui, and O'ahu, and four on the Island of Hawai'i) (Gugger et al., 2018). Despite some shared clusters across islands, these were attributed to recent human-mediated dispersal and limited gene flow was detected between the four oceanic islands of Hawai'i. This also resulted in a score of 2/2 for genetic structure.

In **Scenario (4)**, although four subpopulations could be inferred based on natural distribution, the genetic/genomic studies did not directly support this configuration. Therefore, a score of 0/2 for genetic structure was assigned.

genetic

recorded biological

inferred

Genetic evidence

Is there evidence of genetic structure? ?

Evidence through (multiple can apply): ?

- Population Clustering Analyses
- Pairwise F_{ST} Values
- Private Alleles

Caution: If there is knowledge of human-mediated/controlled genetic structure or extreme bottleneck events, genetic clustering detected in a species does not automatically indicate the presences of separate subpopulations.

1 Metapopulation	2 Subpopulations	7 Subpopulations	4 Subpopulations
☒ Yes ☐ No ☐ No data	☒ Yes ☐ No ☐ No data	☒ Yes ☐ No ☐ No data	☐ Yes ☒ No ☐ No data

Recorded biological evidence

Is there evidence of natural isolation or subpopulation fragmentation (through migration barriers)? ?

Evidence through (multiple can apply): ?

- Occupation of different biogeographical zones
- Occupation of discrete remnants of historical habitats with little/no natural migration

b. Recorded biological evidence

In **Scenario (1)** (null hypothesis), depicting a single metapopulation. For this scenario scoring is reserved to keep the questions identical across all scenarios. The existing evidence did not indicate a single metapopulation. Therefore, a score of 0/2 for genetic structure was assigned.

In **Scenarios (2) and (3)**, despite the presence of genetic structuring, the species occurs naturally across the four main Hawaiian islands (Island of Hawai'i, Maui, O'ahu, and Kaua'i). The observed genetic patterns did not clearly correspond to identifiable migration barriers. Therefore, both scenarios received a score of 0/2 for isolation/migration barriers.

In **Scenario (4)**, the natural occurrence pattern showed distinct occupation of different habitats across islands, which can be interpreted as biogeographical separation and potential migration barriers. This resulted in a score of 2/2 for isolation/migration barriers.

Recorded biological evidence

Is there evidence of natural isolation or subpopulation fragmentation (through migration barriers)?

Evidence through (multiple can apply):

- Occupation of different biogeographical zones
- Occupation of discrete remnants of historical habitats with little/no natural migration

Caution: In this framework, the natural fragmentation of subpopulations should be distinguished from human-mediated/controlled fragmentation of subpopulations, since the latter does not automatically indicate the presences of separate subpopulations.

1 Metapopulation	2 Subpopulations	7 Subpopulations	4 Subpopulations
☒ Yes	☐ Yes	☐ Yes	☒ Yes
☐ No	☒ No	☒ No	☐ No
☐ No data	☐ No data	☐ No data	☐ No data

Inferred evidence

Is there inferred evidence of likely subpopulation fragmentation?

Note: Only relevant if genetic or recorded evidence form the focal species is absent. Evidence of inferred patterns can be used. If nothing is know about the focal species, co-occurring or closely related species could give indication of subpopulation structure.

c. Inferred evidence

In **Scenario (1)** (null hypothesis), depicting a single metapopulation. For this scenario scoring is reserved to keep the questions identical across all scenarios. No inferred evidence was found to support the null hypothesis of a single metapopulation. Therefore, this line of evidence was marked as 'No data' and score 0/2 points.

In **Scenarios (2), (3), and (4)**, no inferred evidence was found to support subpopulation fragmentation. As a result, each scenario received a score of 0/2 in this category, and the evidence was marked as ‘No data’.

Inferred evidence

Is there inferred evidence of likely subpopulation fragmentation?

Note: Only relevant if genetic or recorded evidence from the focal species is absent. Evidence of inferred patterns can be used. If nothing is known about the focal species, co-occurring or closely related species could give indication of subpopulation structure.

Evidence through (multiple can apply):

- **Inferred** subpopulation disjunction (e.g., species distribution modeling)
- Phylogeographic and/or biogeographic evidence **from co-occurring species**
- Assumption of biogeographical data/refugia **without direct evidence**
- **Inferred** dispersal distance and buffer according to the CBD indicators

1 Metapopulation	2 Subpopulations	7 Subpopulations	4 Subpopulations
☐ Yes	☐ Yes	☐ Yes	☐ Yes
☐ No	☐ No	☐ No	☐ No
☒ No data	☒ No data	☒ No data	☒ No data

3. Provide name and e-mail address (optional). This will allow the results to be sent to your email address.

Notes

Name & Surname

Jane Doe

Email Address

jane.doe@xxxx

Data protection disclaimer

By providing your name and email, you will receive the results in your inbox, as well as on the next page and may receive communications from us to participate further in the development of this framework. If you **choose not to enter your email**, your results will still be presented to you on the next page.

Submit Data

4. Click the **“Submit Data” button** to receive the results (mandatory to get results).

5. Interpreting the results.

This is the results page. It lists each question and the responses entered by the user as verification. Below the questions is a summary of the scores for each scenario.

For the Hawaiian koa tree, three of the four candidate scenarios (**Scenarios (2), (3), and (4)**) meet the minimum threshold of two points, and therefore qualify as subpopulation scenarios. Where scores are tied, we recommend selecting the one scenario which is based on the most robust data to maximize the conservation of genetic diversity. In this case, Scenario (3), based on high-resolution genomic markers (11,000 SNPs from >300 samples via genotyping-by-sequencing, Gugger et al., 2018), identified seven distinct subpopulations, while using nine microsatellites from 170 samples (Fredua-Agyeman et al., 2008), identified only two subpopulations. Although microsatellites remain valuable in population genetics, the two-subpopulation scenario (2) is unlikely to represent the true subpopulation structure. Instead, it might better reflect higher-level structuring at the ESU level, along with Scenario (4) (4 islands subpopulations). Overall, the seven-subpopulation scenario (Scenario 3) better represents subpopulation structure.

Scenarios (2), (3), and (4) scored equally higher (two points) and therefore can be tested in Phase 2.

Phase 1: Identify Subpopulations

This survey is best views on Desktop, or rotate your phone for a better view.

Please use the dropdown menu below for additional information.

- Subpopulations
- Evolutionarily Significant Units (ESU)
- Possible Evolutionarily Significant Unit (pESU)
- Gene Flow
- Potential IUCN Species assessment
- References

Scoring:

- 2 points or higher = distinct subpopulations
- Less than 2 points = stop assessment

Lines of evidence

genetic
recorded biological

Thank you Jane Doe, for participating in the **CGSG Subpopulation and ESU survey**.

The species you tested is the **Acacia koa**

Your Survey Results:

1. Is there evidence of population structure? (2 points)

1 Metapopulation Yes	2 Subpopulations Yes	7 Subpopulations Yes	4 Subpopulations No
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2. Is there evidence of natural isolation or subpopulation fragmentation (through migration barriers)? (2 points)

1 Metapopulation Yes	2 Subpopulations No	7 Subpopulations No	4 Subpopulations Yes
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3. Is there inferred evidence of likely subpopulation fragmentation? (1 points)

1 Metapopulation No data	2 Subpopulations No data	7 Subpopulations No data	4 Subpopulations No data
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Total score for the scenario: 1 **Metapopulation** is 0
Total score for the scenario: 2 **Subpopulations** is 2
Total score for the scenario: 7 **Subpopulations** is 2
Total score for the scenario: 4 **Subpopulations** is 2

For scenarios that scored 2 points or higher, please use the button below to continue to Phase 2.

GO TO PHASE 2 >>

You'll be receiving the results in your inbox soon!

6. Click on the “Phase 2” button to proceed.

7. As in Phase 1, enter the species and scenario information. This does not get carried over from Phase 1.

Based on the results of Phase 1, we tested four candidate scenarios for delineating Evolutionarily Significant Units (ESUs): (1) a single ESU (2) two ESUs (3) seven ESUs and (4) four ESUs.

Phase 2: Delineate Evolutionarily Significant Units (ESUs)

Please use the information provided by the email sent, to complete the following ESU survey.

Please use the dropdown menu below for additional information.

- Subpopulations
- Evolutionarily Significant Units (ESU)
- Probable Evolutionarily Significant Unit (pESU)
- Gene Flow
- Potential IUCN Species assessment
- References

Scoring:

- 10 points or higher = distinct ESU
- Below 10 points:
 - a) 10 points cannot be reached or exceeded (even if all missing data were supportive of the tested scenario) = **no support for the ESU - Scenario.**
 - b) 10 points are currently not reached or exceeded, but could if missing data subsequently prove to be supportive of the tested scenario = **scenario indicates possible ESUs (pESUs)***

Which species are you testing?

Acacia koa

How to set up scenarios to test:
This survey allows the evaluation of **multiple ESU scenarios**.

The **first scenario always represents the null hypothesis**, which assumes a **single Evolutionarily Significant Unit (ESU)**. This one-ESU model must be tested first and serves as the baseline against which all alternative scenarios involving multiple ESUs are compared.

When selecting scenarios that include **multiple ESUs** (i.e. not the default single ESU), **all ESUs must be mutually distinct and supported by evidence**. Each ESU must have a clear evidentiary basis for inclusion in the scenario being tested. If any two ESUs cannot be distinguished from one another, they should be merged, and a revised scenario should be considered for testing.

How many scenarios are you testing?

4 Scenarios

1 ESU <i>Scenario 1 name</i>	2 ESUs <i>Scenario 2 name</i>	7 ESUs <i>Scenario 3 name</i>	4 ESUs <i>Scenario 4 name</i>
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Genetic evidence

8. Answer the questions for each line of evidence for each scenario being investigated.

a. Genetic evidence

In **Scenario (1)** (null hypothesis), depicting a single ESU, scoring was reserved to maintain consistency across scenarios. A cytology study found tetraploidy ($2n=52$) across all islands (Shi, 2003) resulting in 6/6 for karyotype variation. However, no evidence for evolutionary distinctiveness was found (0/6). Landscape genomics showed divergence correlated with precipitation (Gugger et al., 2018) and island-based seed zones are supported (Dudley et al., 2017), contradicting a single ESU and resulting in 0/6 for adaptive divergence.

In **Scenarios (2)**, based on two ESUs, tetraploidy across islands (Shi, 2003) resulted in 0/6 for karyotype variation. No evidence for evolutionary distinctiveness was found (0/6). Landscape genomics did not support adaptation patterns matching two ESUs (Gugger et al., 2018), resulting in 0/6 for adaptive divergence.

In **Scenario (3)**, based on seven ESUs, tetraploidy across islands (Shi, 2003) resulted in 0/6 for karyotype variation. Admixture among subpopulations (Gugger et al. 2018) contradicts ESU criteria of restricted gene flow (leading to unique evolutionary trajectories), resulting in

a score of 0/6 for evolutionary distinctiveness. No adaptive divergence matching this scenario was found (0/6) (Gugger et al., 2018).

In **Scenario (4)**, based on four ESUs corresponding to the main Hawaiian islands, tetraploidy across islands (Shi, 2003) resulted in 0/6 for karyotype variation. No evidence for evolutionary distinctiveness was found (0/6). Genetic divergence correlated with precipitation (Gugger et al., 2018), and island-based seed zones are supported (Dudley et al., 2017), resulting in 6/6 for adaptive divergence.

prove to be supportive of the tested scenario = scenario indicates possible ESUs (pESUs)*

*However, if a candidate scenario currently only scores in recorded biological evidence-types (so non-genetic), the scenario should be regarded as 'data-deficient' rather than as a pESU scenario.

Lines of evidence

genetic

→

recorded biological

→

inferred

Genetic evidence

Is there inherited variation in chromosome numbers, ploidy level or chromosome structure?

Evidence through (multiple can apply): i

- Karyotype variation
- Difference in ploidy levels between units
- Structural variation of chromosomes between units

1 ESU	2 ESUs	7 ESUs	4 ESUs
☐ Yes	☐ Yes	☐ Yes	☐ Yes
☒ No	☒ No	☒ No	☒ No
☐ No data	☐ No data	☐ No data	☐ No data

Is there evidence of long-term reproductive isolation?

Evidence through (multiple can apply): i

- Reciprocal Monophyly
- Molecular Estimate of Divergence Time

1 ESU	2 ESUs	7 ESUs	4 ESUs
☐ Yes	☐ Yes	☐ Yes	☐ Yes
☐ No	☐ No	☐ No	☐ No
☒ No data	☒ No data	☒ No data	☒ No data

Is there evidence of local adaptation or adaptive divergence?

Evidence through (multiple can apply): i

- Genomic signals of local adaptation
- Evidence from hybrid zones

1 ESU	2 ESUs	7 ESUs	4 ESUs
☒ Yes	☐ Yes	☐ Yes	☒ Yes
☐ No	☒ No	☒ No	☐ No
☐ No data	☐ No data	☐ No data	☐ No data

b) Recorded biological evidence

In **Scenario (1)** (null hypothesis), depicting a single ESU, provenance tests revealed inherited phenotypic differences among islands (Baker et al., 2009), yielding 0/4 for inherited variation. No evidence was found for recorded variation or traditional knowledge (0/2 each). In **Scenarios (2) and (3)**, provenance tests revealed variation at the island level rather than matching the proposed ESU structure (Baker et al., 2009), resulting in 0/4 for inherited variation. No evidence was found for recorded variation or traditional knowledge in either scenario (0/2 each).



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10. Click the “**Submit Data**” button to receive the results.

11. Interpreting the results.

As with the subpopulation survey results, each question and the answers provided are presented on the results page. A summary of the total scores for each scenario is presented below that.

Only scenario (4) (four ESUs) scored 10 points out of 16 points evaluated (with available data), providing strong support for four distinct ESUs. This score combines one genetic line of evidence (‘Adaptive divergence’) with one recorded line of evidence (‘Inherited characteristic variation’). Scenario (1) received 6 points out of 16 evaluated and scenario (2) (two ESUs) and scenario (3) (seven ESUs) received zero points due to lack of supporting evidence.

Phase 2: Delineate Evolutionarily Significant Units (ESUs)

Please use the information provided by the email sent, to complete the following ESU survey.

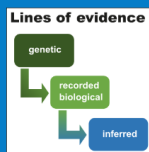
Please use the dropdown menu below for additional information.

- Subpopulations
- Evolutionarily Significant Units (ESU)
- Probable Evolutionarily Significant Unit (pESU)
- Gene Flow
- Potential IUCN Species assessment
- References

Scoring:

- 10 points or higher = distinct ESU
- Below 10 points:
 - a) 10 points cannot be reached or exceeded (even if all missing data were supportive of the tested scenario) = **no support for the ESU - Scenario.**
 - b) 10 points are currently not reached or exceeded, but could if missing data subsequently prove to be supportive of the tested scenario = **scenario indicates possible ESUs (pESUs)***

*However, if a candidate scenario currently only scores in recorded biological evidence-types (so non-genetic), the scenario should be regarded as ‘data-deficient’ rather than as a pESU scenario.



Thank you Jane Doe, for participating in the **CGSG Subpopulation and ESU survey**.

The species you tested is the **Acacia koa**

Your Survey Results for Phase 2:

1. Is there inherited variation in chromosome numbers, ploidy level or chromosome structure? (6 points)

1 ESU No	2 ESUs No	7 ESUs No	4 ESUs No
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2. Is there evidence of long-term reproductive isolation? (6 points)

1 ESU No data	2 ESUs No data	7 ESUs No data	4 ESUs No data
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3. Is there evidence of local adaptation or adaptive divergence? (6 points)

1 ESU Yes	2 ESUs No	7 ESUs No	4 ESUs Yes
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4. Is there evidence of inherited characteristic differences? (4 points)

1 ESU Yes	2 ESUs No	7 ESUs No	4 ESUs Yes
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5. Is there evidence supporting transmitted characteristic differences? (2 points)

1 ESU No data	2 ESUs No data	7 ESUs No data	4 ESUs No data
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6. Is there traditional knowledge suggesting distinctiveness between units? (2 points)

1 ESU No data	2 ESUs No data	7 ESUs No data	4 ESUs No data
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Total score for the scenario: **1 ESU** is 6
Total score for the scenario: **2 ESUs** is 0
Total score for the scenario: **7 ESUs** is 0
Total score for the scenario: **4 ESUs** is 10

Scenario scoring:

- Scenarios scoring 10 points or higher = distinct ESUs
- Scenarios scoring below 10 points:
 - without potential to reach/exceed 10 points = no support for tested scenario
 - with potential to reach/exceed 10 points = possible ESU (pESU) status

You'll be receiving the results in your inbox soon!

You've reached the end of the survey. We trust that the information will provide insight into the management of your species.

References

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